

Joint Attention-Field × PRF model

Distractor-driven shifts of population receptive
fields in human visual cortex

meeting with Dock Duncan & Jan Theeuwes

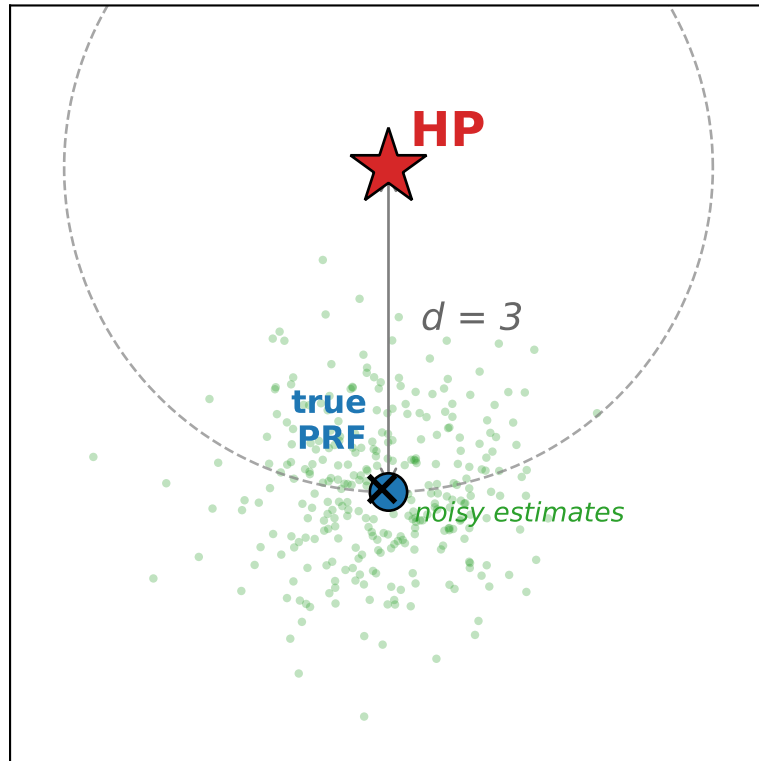
retsupp · 7T fMRI · 30 subjects · 4 HP-distractor blocks · sweeping bar PRF mapping

(a) Distance binning is biased

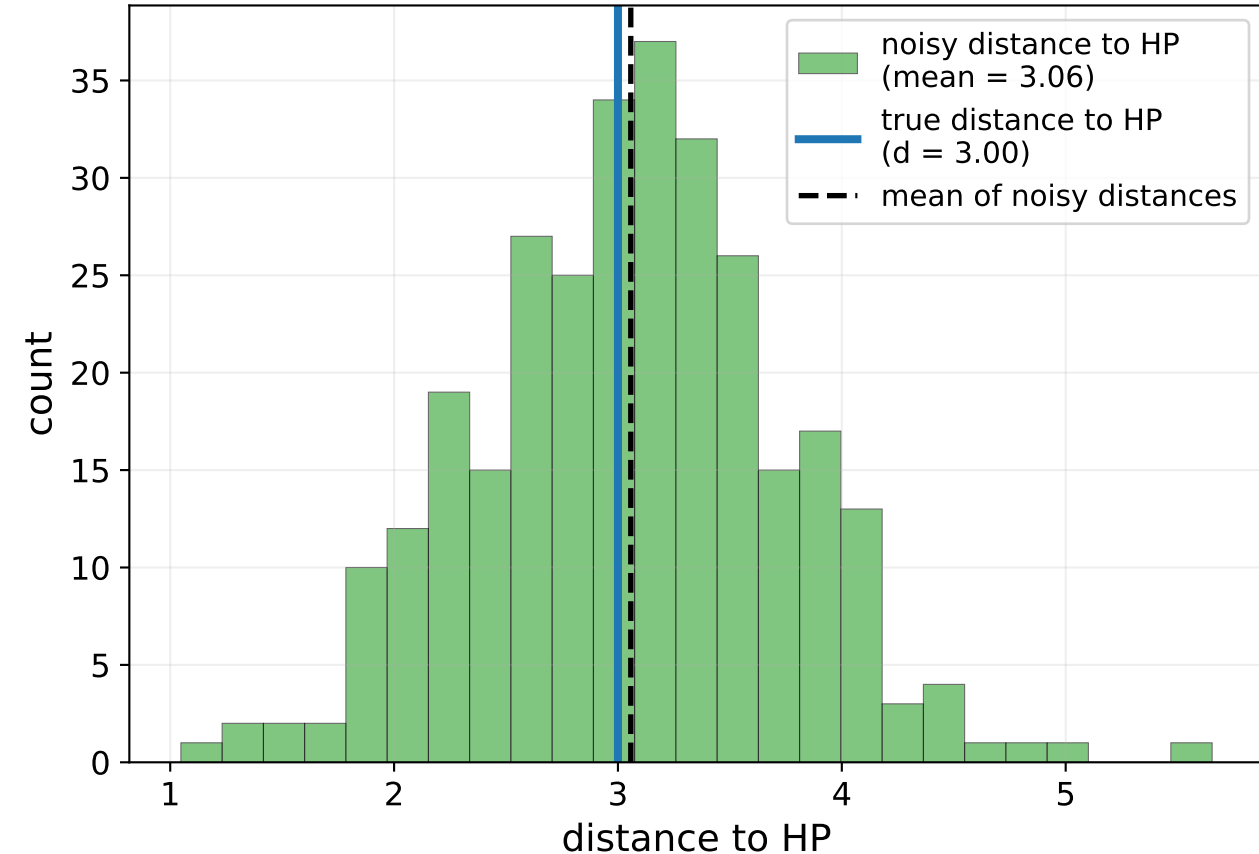
Random noise on PRF position pushes distance-to-HP up.

**(a) Random noise pushes "distance to HP" upward.
Hence "the PRF shifted AWAY from HP" can be a noise artifact.**

Take any voxel a distance d from HP.
Add isotropic noise to its PRF estimate.



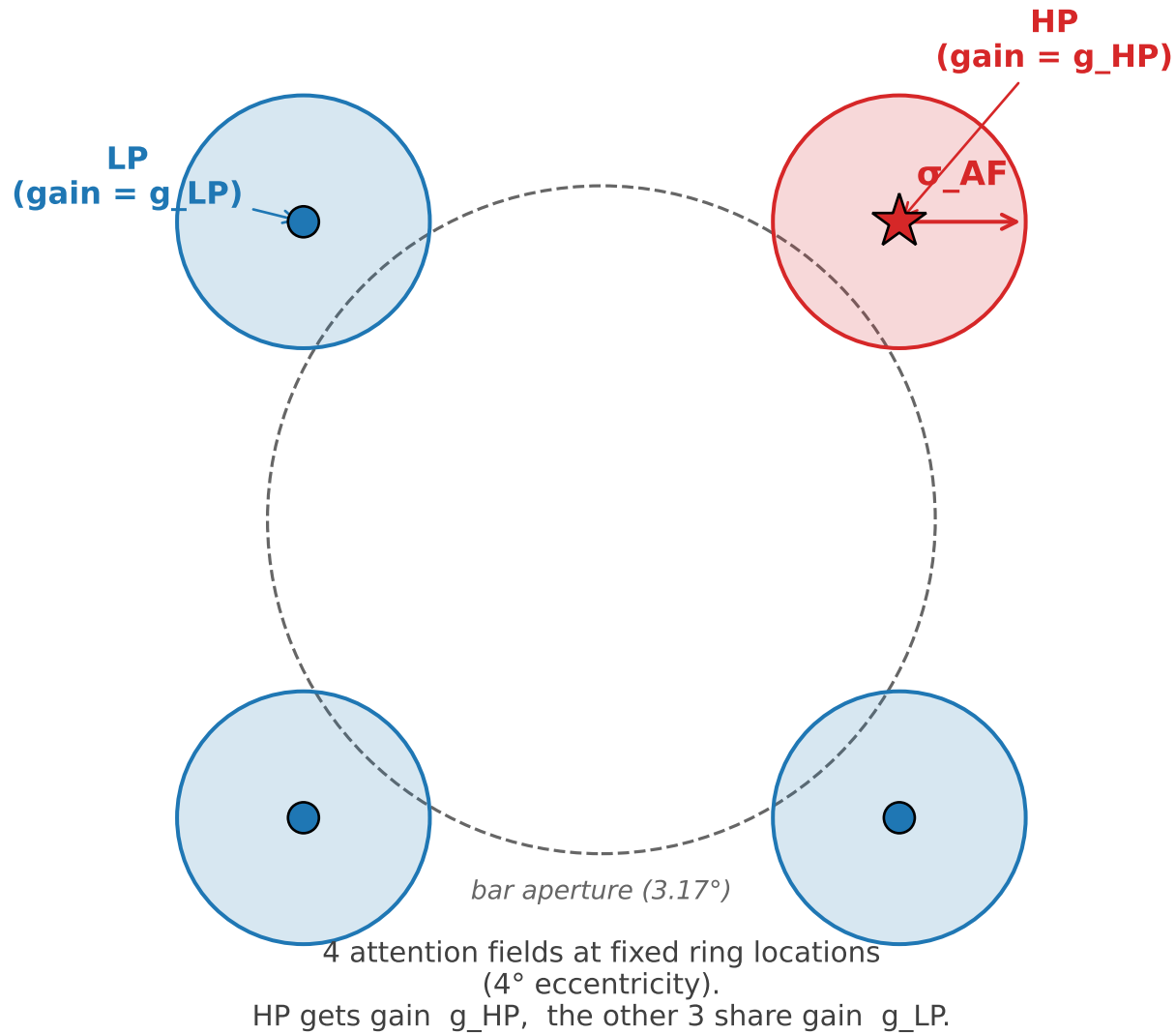
On AVERAGE, noisy distances $>$ true distance.
(Distance is convex — Jensen.)



(b) The model

4 attention fields $\times$ Gaussian voxel kernel; 3 shared parameters per ROI per subject.

(b) The model



Forward model

For each voxel v under HP condition C :

$$R_{v_C}(g) = M_C(g) \cdot S_v(g)$$

with the modulation field

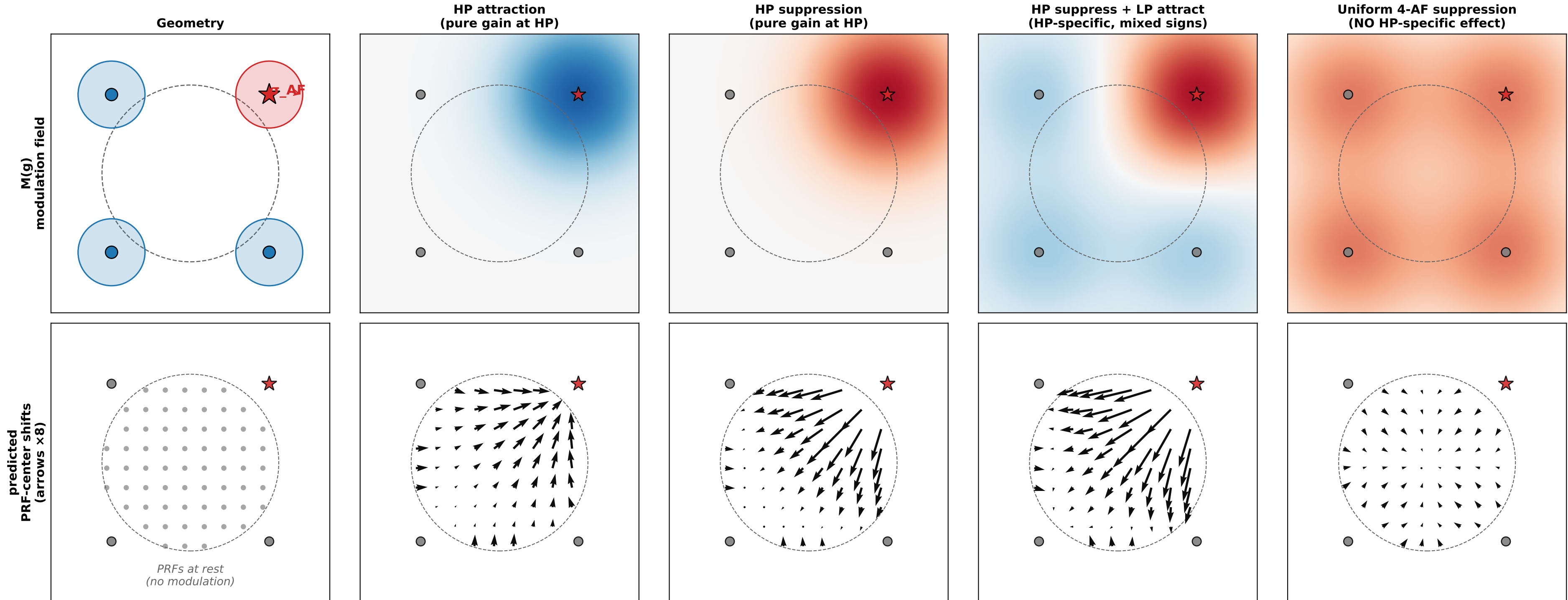
$$M_C(g) = 1 + g_{HP} \cdot A_{HP_C}(g) + g_{LP} \cdot \sum A_{LP}(g)$$

3 SHARED parameters per ROI:

σ_{AF}	AF size
g_{HP}	gain at HP location
g_{LP}	gain at LP locations

Negative gain $\Rightarrow$ suppression.

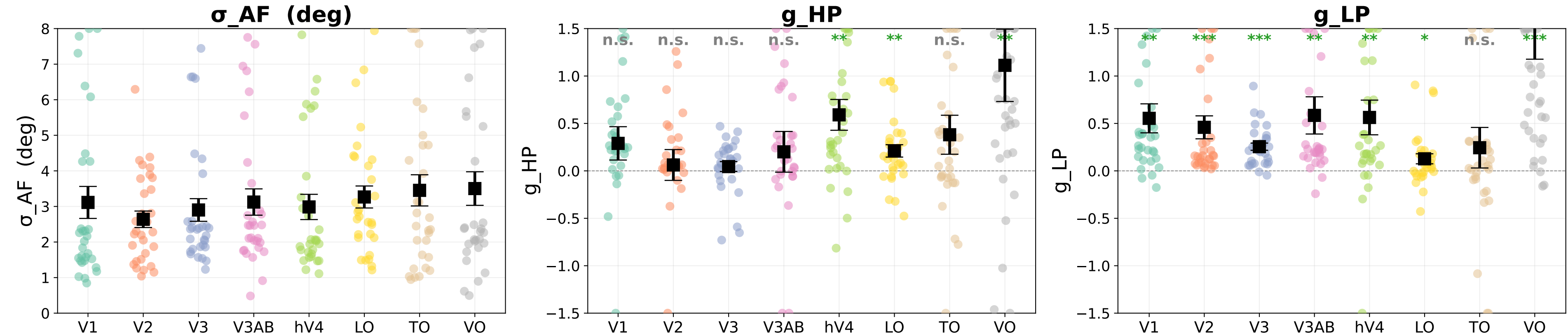
(b) What the model predicts at different parameter settings.
Top: modulation field $M(g)$ (RED = suppression, BLUE = attraction). **Bottom: predicted PRF-center shifts under $HP=\star$.**



(c) Parameter estimates

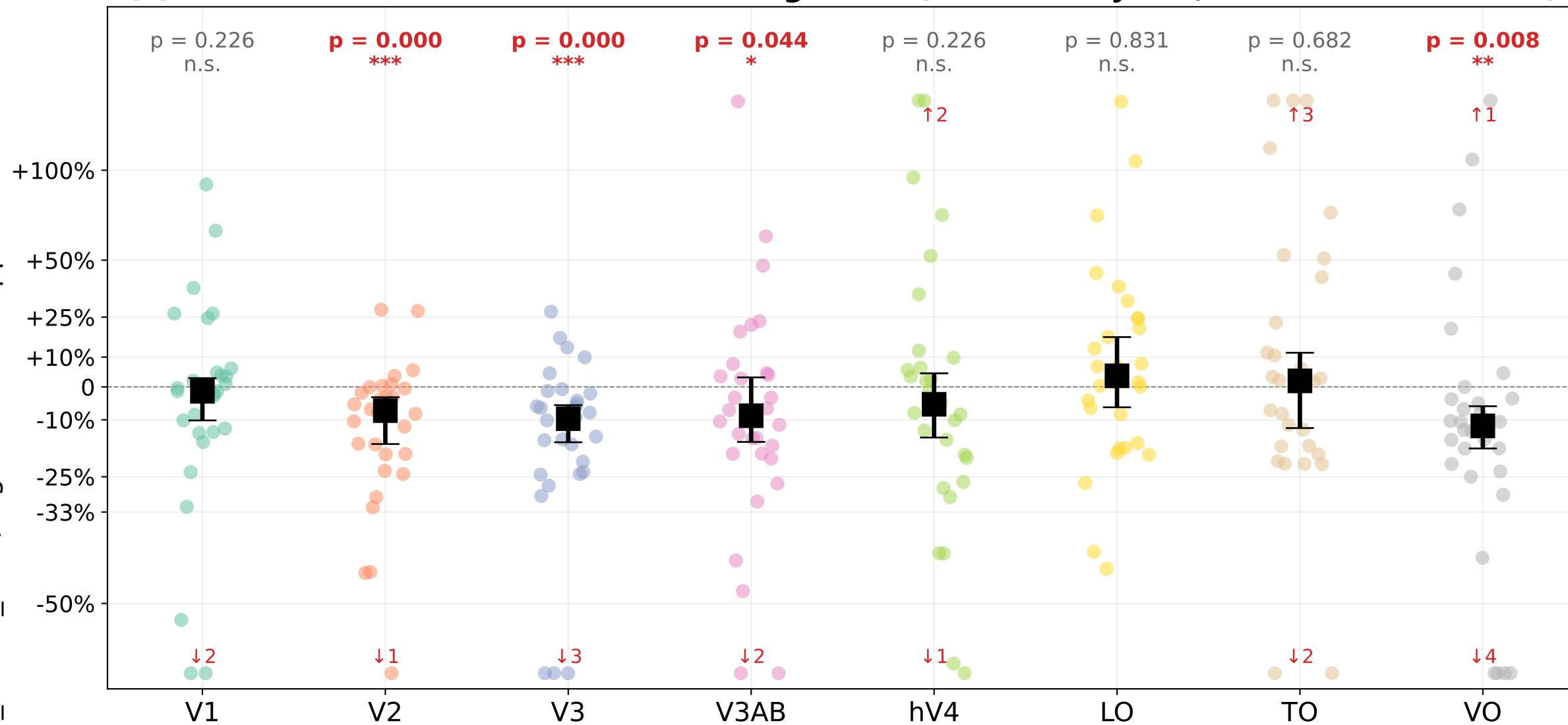
Per-ROI distributions and HP-vs-LP test ($N = 30$ subjects).

(c) AF parameter estimates per ROI (30 subjects, 240 fits) — dots = subjects, black square = mean $\pm$ SEM

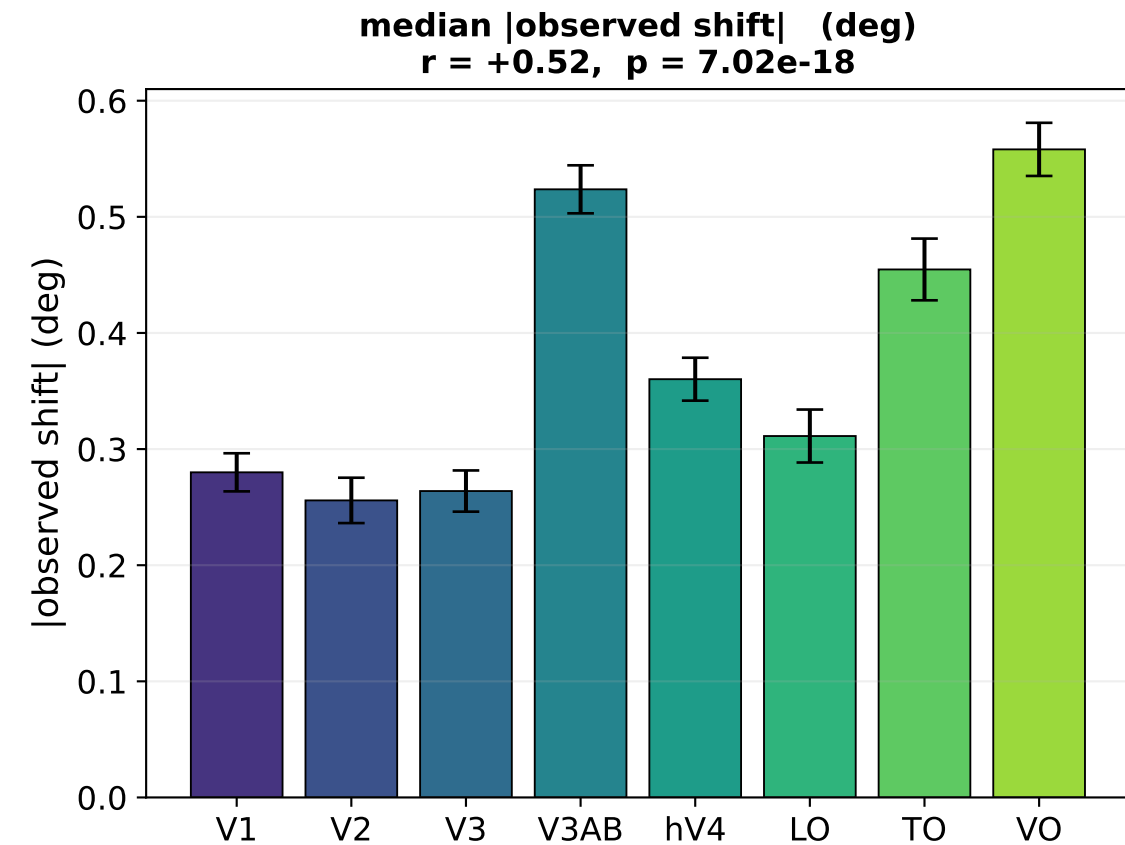
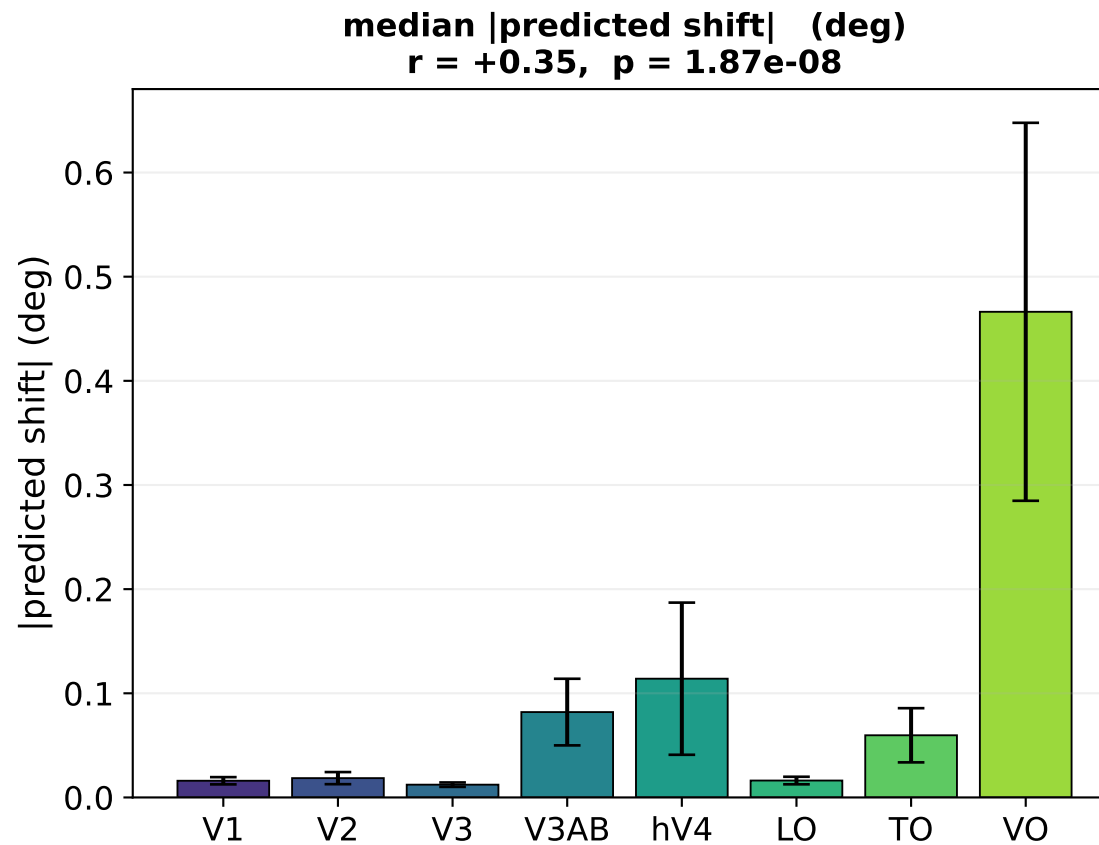
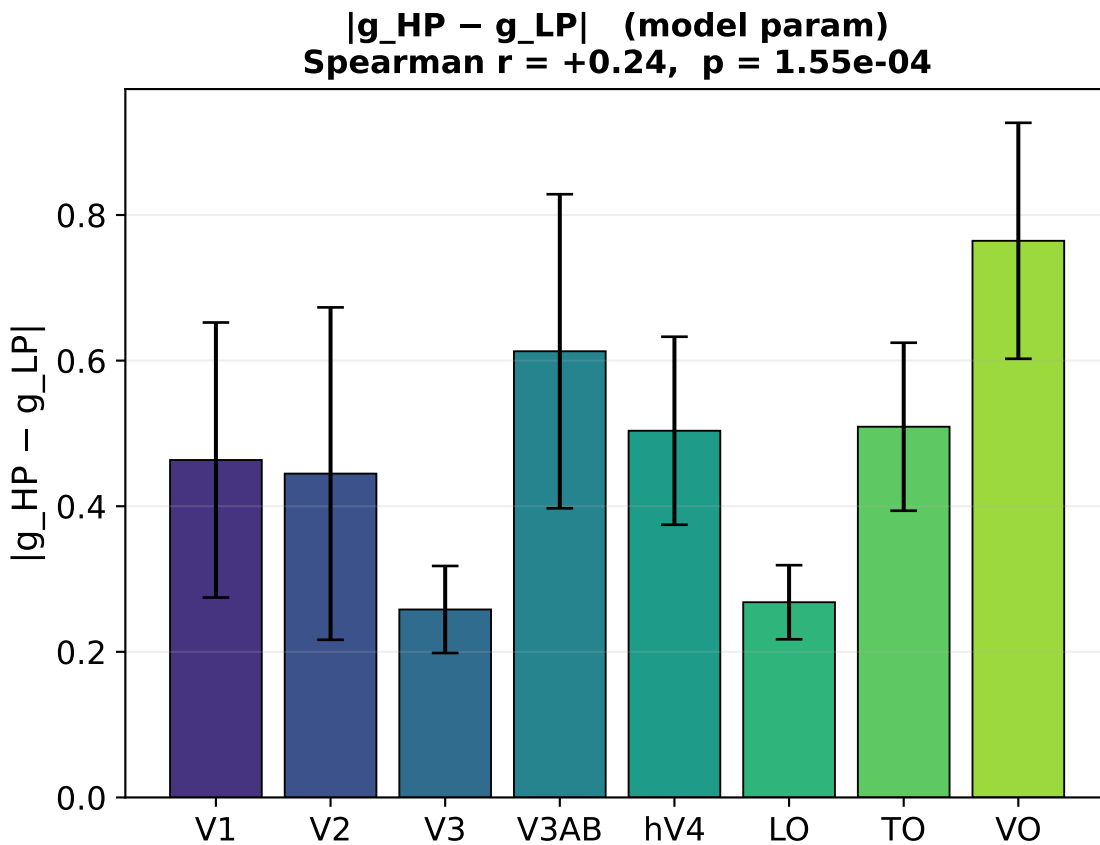


M_{HP} / M_{LP} (negative $\Rightarrow$ HP suppressed more than LP)

(c) HP-vs-LP test — scale-invariant log-ratio (N = 30 subjects, Wilcoxon one-sided)



(d'') Shifts grow up the visual hierarchy.
V1 → V2 → V3 → V3AB → hV4 → LO → TO → VO (mean across 30 subjects ± SEM).

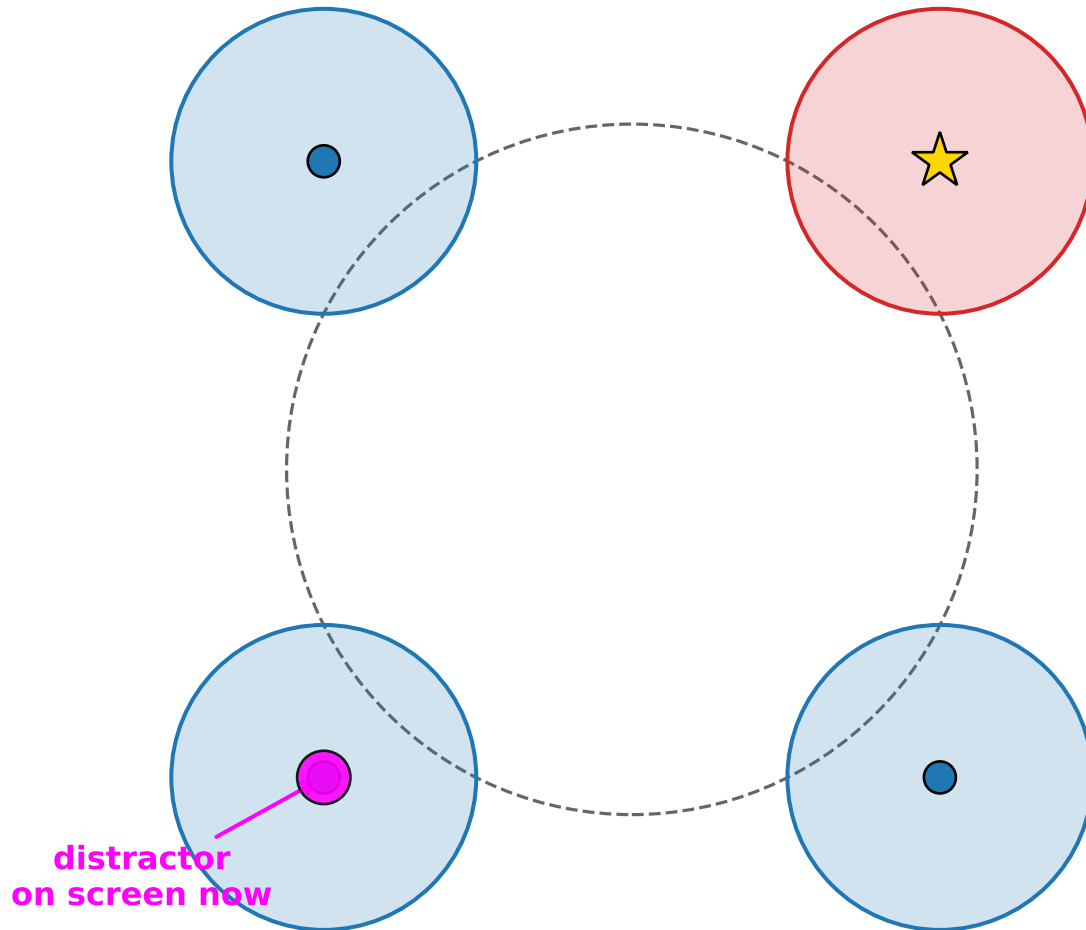


ic AF — distractor pulses on top of the sustained p

5 shared parameters per ROI; same per-voxel Gaussian kernel.

(e) Dynamic AF — adds a per-TR distractor pulse

sustained AF at all 4 ring locations
+ TRANSIENT AF at the location currently
showing a distractor (on/off per TR)



Five shared parameters per ROI

σ_{AF} AF size (shared sustained $\leftrightarrow$ dynamic)

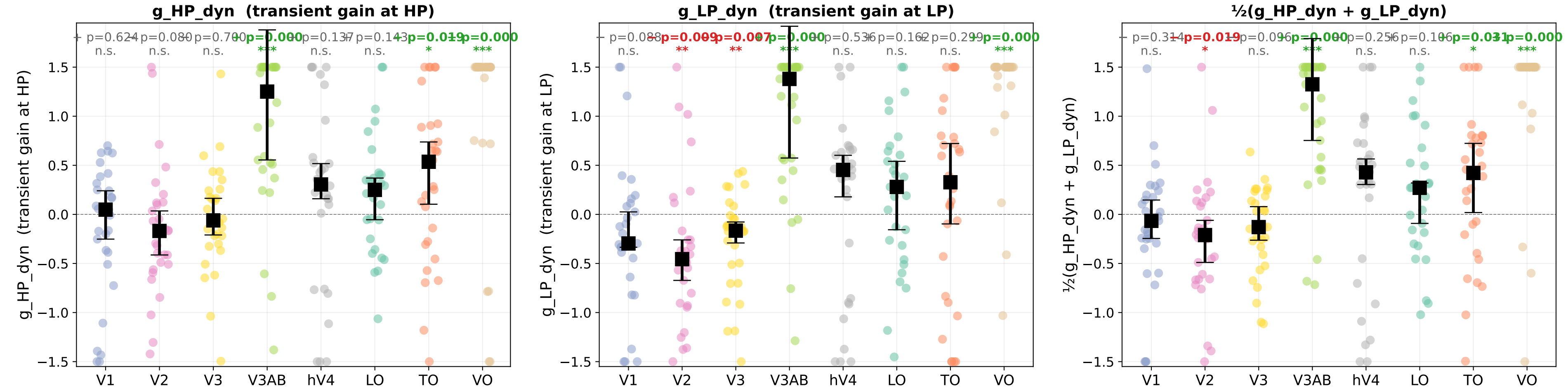
g_{HP} sustained gain at HP
 g_{LP} sustained gain at LP

g_{HP_dyn} transient gain when distractor at HP
 g_{LP_dyn} transient gain when distractor at LP

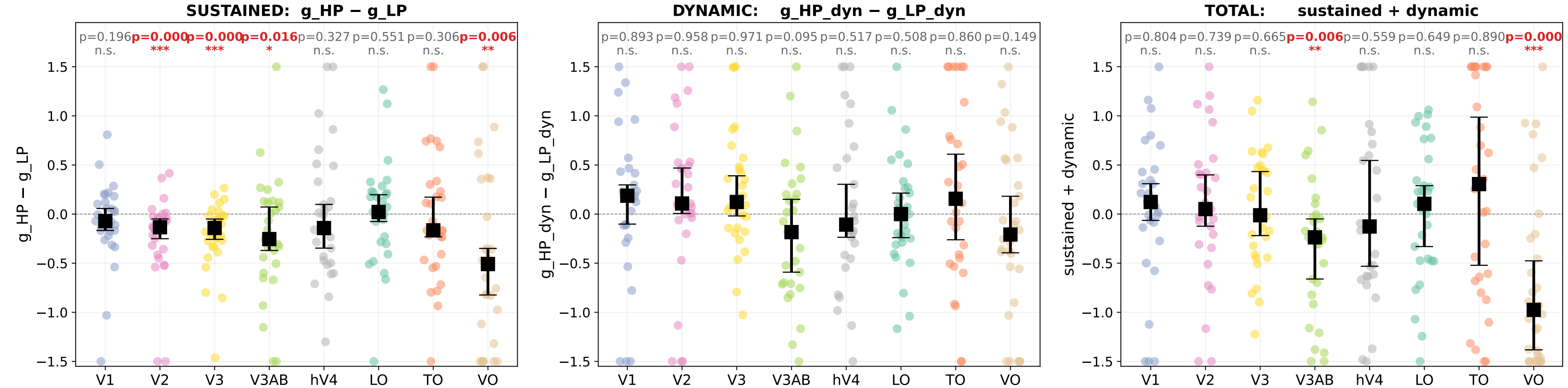
$$M(g, t) = 1 + g_{HP} \cdot A_{HP_C}(g) + g_{LP} \cdot \sum A_{LP}(g) + g_{HP_dyn} \cdot d_{HP}(t) \cdot A_{HP_C}(g) + g_{LP_dyn} \cdot \sum d_{LP}(t) \cdot A_{LP}(g)$$

$d_{\square}(t) \in [0, 1]$: fraction of TR with distractor on screen at ring location $\square$.

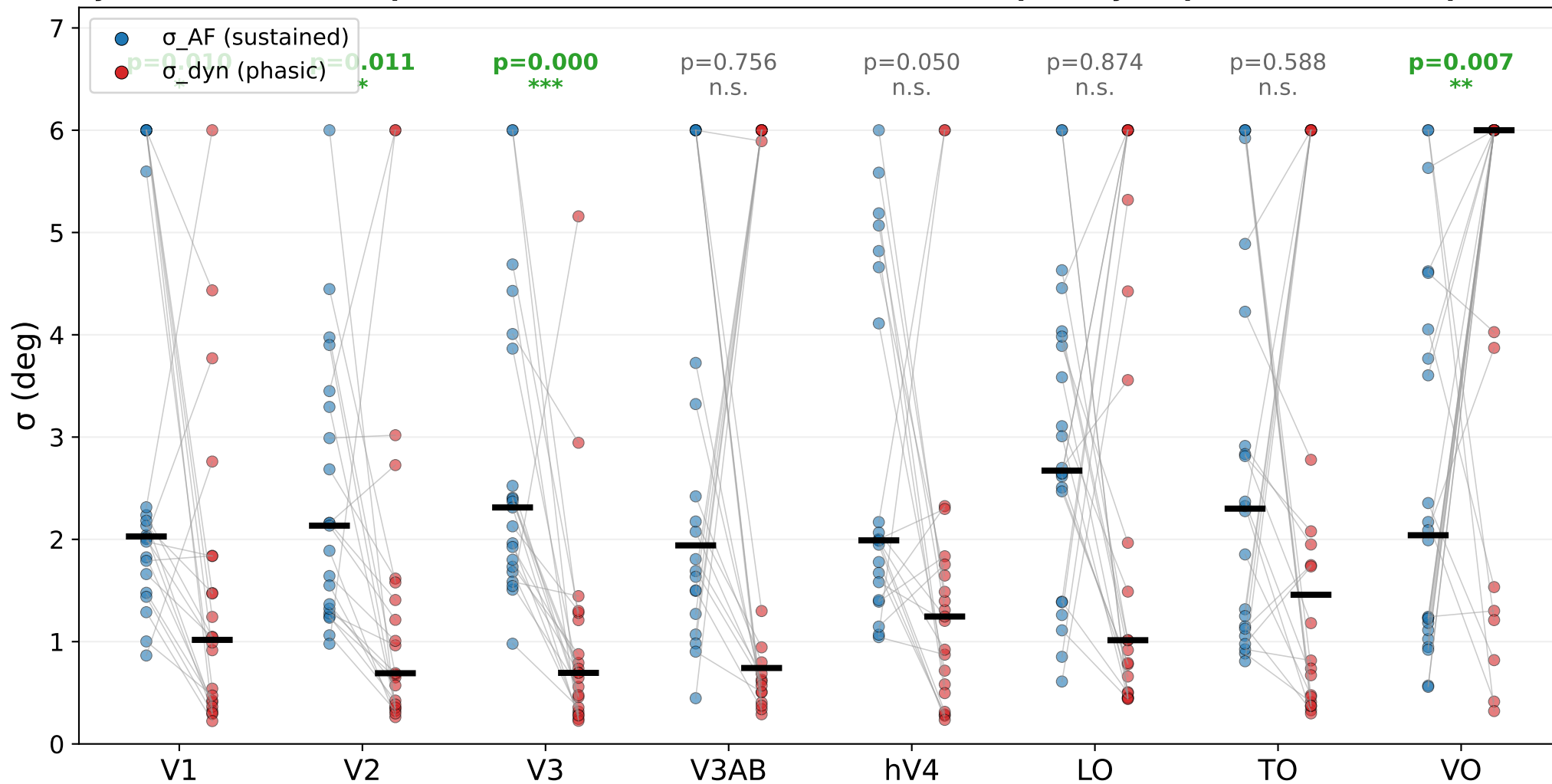
(e) Is there transient attention at all? g_{HP_dyn} and g_{LP_dyn} vs zero per ROI.
GREEN = significant CAPTURE (attraction). RED = significant SUPPRESSION.



(e) HP–LP differential: sustained, dynamic, and total per ROI.
Wilcoxon one-sided H_1 : HP < LP. V2/V3 cancel — dynamic offsets sustained. V3AB/VO add.



σ_{AF} (sustained, blue) vs σ_{dyn} (phasic distractor, red) — v3 model, per-fit paired comparison
In early visual cortex, the phasic AF is narrower than the sustained priority map → two distinct spatial scales.

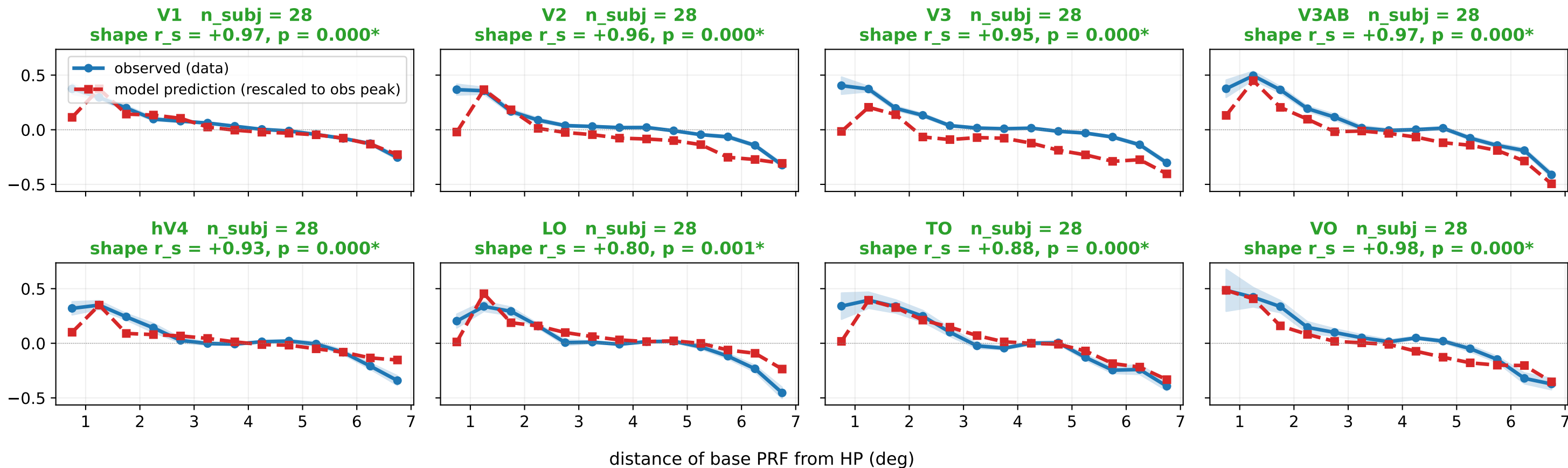


(g) Observed vs predicted shifts

Per-ROI projection-vs-distance from HP. Solid = data, dashed = model.

(g) Qualitative pattern test: model SHAPE vs data SHAPE.
Per ROI: projection on away-from-HP axis vs distance from HP.
Predicted curve rescaled to match observed peak — magnitude is confounded by Jensen & kernel mismatch, but the SHAPE (peak location, decay) is the test.

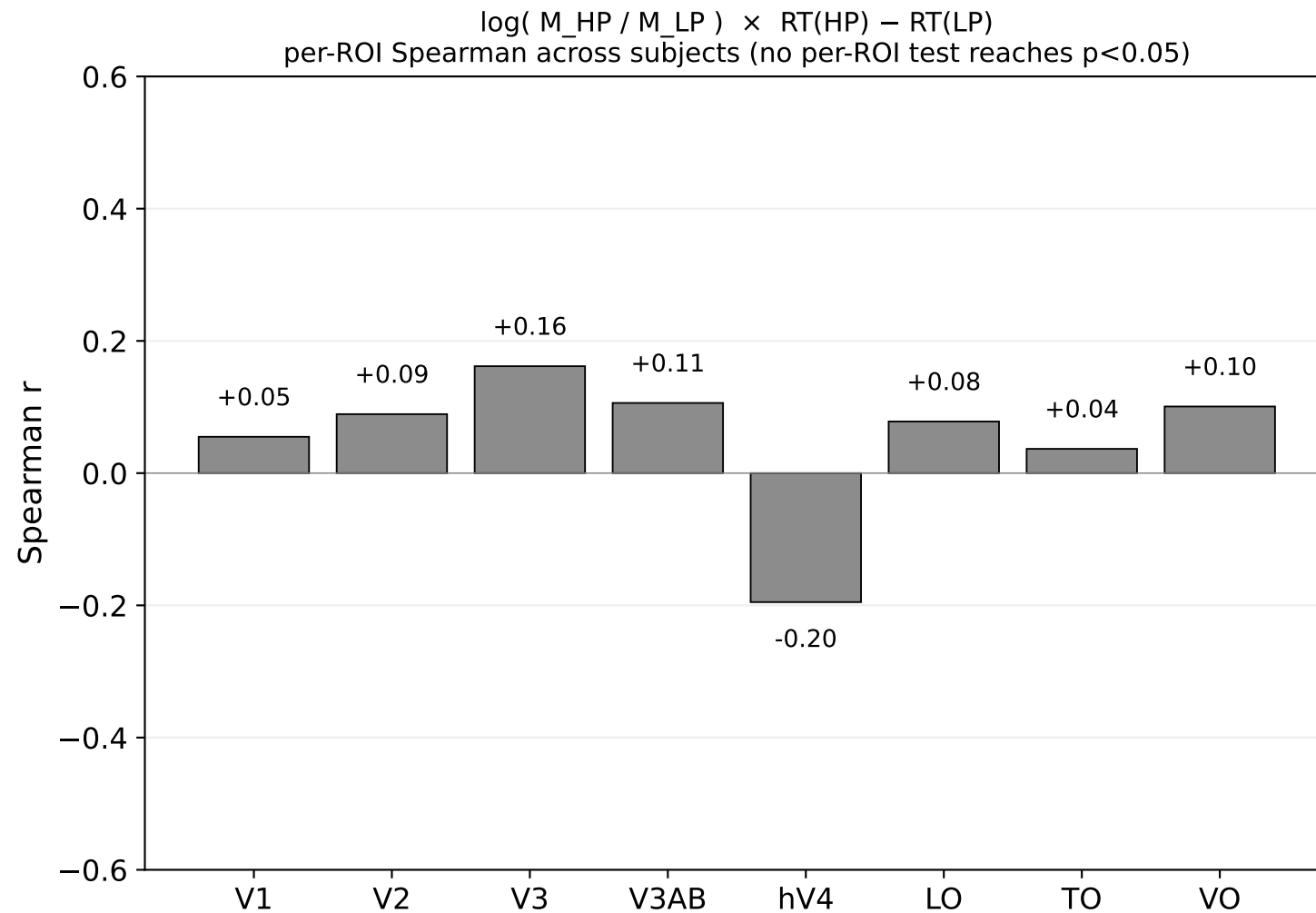
projection on away-from-HP axis (deg)



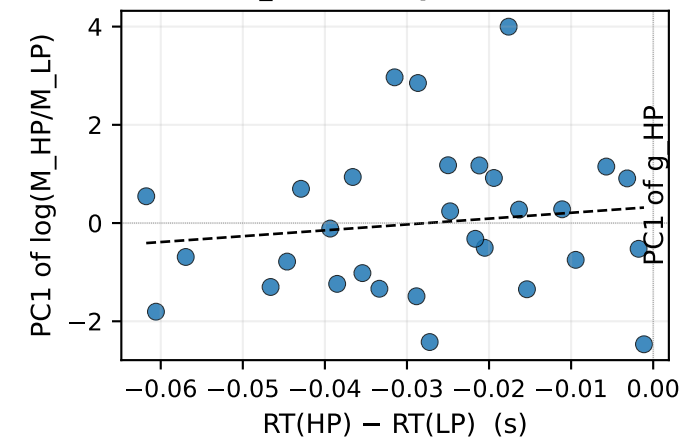
(f) Correlations with behavior

Subject-level co-variation between AF parameters and distractor RT cost.

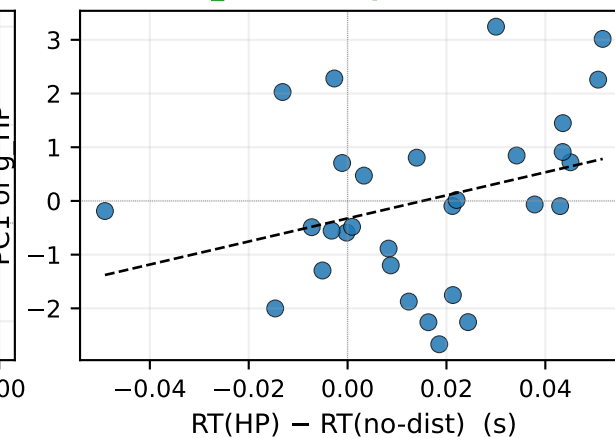
(f) Correlations with behavior (N = 30 subjects)



targeted HP-suppression test
 $r_s = +0.15$, $p = 0.439$



overall HP-AF gain $\times$ HP cost
 $r_s = +0.37$, $p = 0.046^*$



PCA across ROIs collapses 8 weak per-ROI signals into one subject score. The TARGETED test (HP-vs-LP $\times$ HP-vs-LP) is null with $N=30$. The omnibus $g_{HP} \times$ HP-cost test reaches *.

Summary

1. Naive distance-from-HP analyses are biased by Jensen / noise.
The pattern, not the magnitude, is what tests the model.
2. Joint AF + PRF model (4 ring AFs, AF+ formulation, single Gaussian voxel kernel) fits per (subject, ROI) at the BOLD level.
3. HP-specific suppression survives in early-to-ventral cortex:

V2	Wilcoxon $p = 0.0002$	80% of 30 subjects $g_{HP} < g_{LP}$
V3	Wilcoxon $p = 0.0002$	80%
V3AB	Wilcoxon $p = 0.05$	
V0	Wilcoxon $p = 0.003$	80%
4. Per-ROI predicted shift vector fields show what each area is doing – see (d) for the spatial pattern at each ROI's empirical median parameters.

Caveats:

- Model still has identifiability issues ($\sigma_{AF} \times |gain|$ trade-off in mid-tier ROIs) but the rank-based HP-vs-LP test survives.
- PRED side uses single Gaussian voxel kernels; OBS side uses DoG (model 4). Apples-to-apples Gaussian conditionwise refit is in flight on cluster.